MILITARY SPECIFICATION SHEET

CABLE ASSEMBLIES, NONPRESSURE PROOF, FIBER OPTIC, METRIC

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-49292.

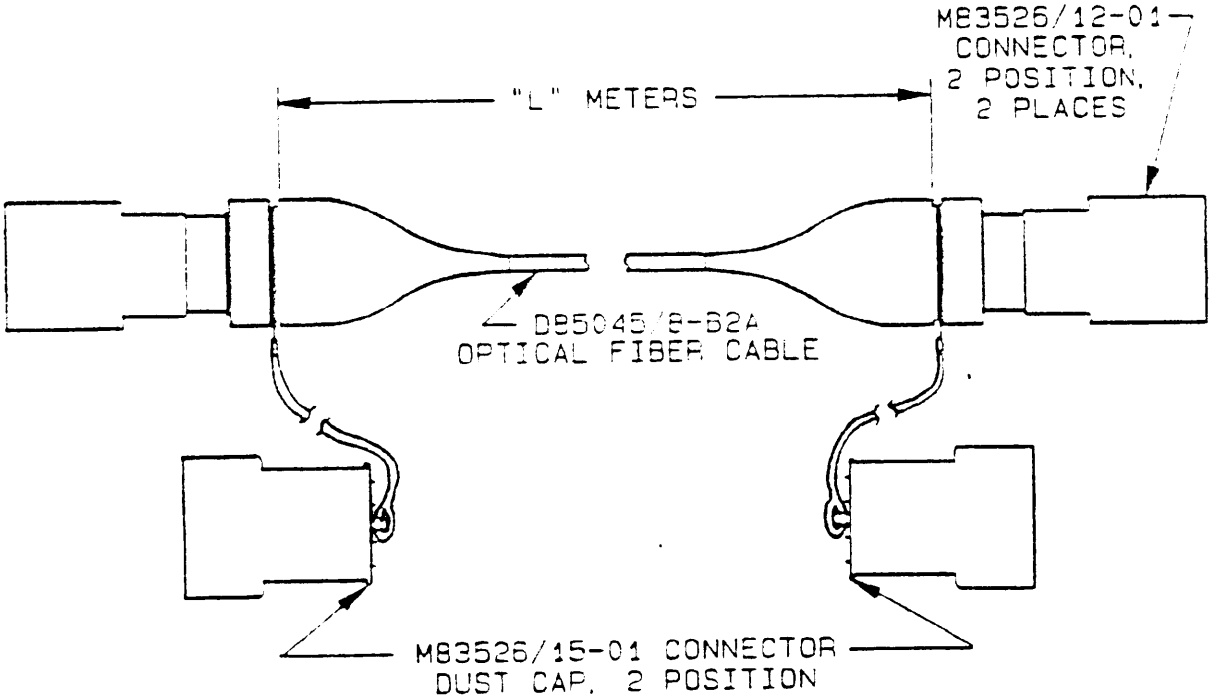


FIGURE 1. Outline drawing, cable assembly (hermaphroditic) fiber optic.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Color: Colors for coding of fibers within the cable assembly shall be blue and orange (see table I).

TABLE I. Color-coded termination configuration.

Termini	Buffer color	Termini
1	Blue	2
2	Orange	1

Component parts:

Connectors shall be in accordance with the following:

M83526/12-01, two required.

Dust cover assembly: M83526/15-01, two required.

Cable: D85045/8-B2A.

Assembly: In accordance with figure 1, and tables I and II.

Length: In accordance with table II.

Weight: In accordance with table II.

Performance:

Optical requirements:

Insertion loss (maximum assembly attenuation): Measurements made at 1290 nm ±10 nm. Test method B1.

Part or Identifying number (PIN)	-46°C	+25°C	+71°C
M49292/4-01	3.25 dB	2.75 dB	2.75 dB
M49292/4-02	2.5 dB	2.0 dB	2.0 dB
M49292/4-03	2.0 dB	1.5 dB	1.5 dB

Optical fiber bandwidth: $\geq$ 400 MHz-km at 1290 nm, ±10 nm.

Cable retention: Not applicable.

Cable seal flexing: Not applicable.

Thermal shock: Not applicable.

Fluid immersion: Not applicable.

Temperature range:

Operating: -46°C to +71°C.

Storage: -57°C to +85°C.

Dust: Not applicable.

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Life test: Not applicable.

Salt fog: Not applicable.

Storage temperature: 240 hours at each extreme for a total of 480 hours.

PIN: PINs shall be in accordance with table II.

TABLE II. PINs.

Military PIN M49292/4-	Length "L" (meters)	Weight (kg) maximum	Fiber size µm
01	1,010 ±10	34.0	50/125
02	310 ±10	12.7	50/125
03	1.0 ±0.1	2.3	50/125

Qualification inspection shall consist of performing the inspections and optical tests specified in table III herein, in the sequence shown.

Group A inspection shall consist of performing the inspections and optical tests specified in table IV herein, in the sequence shown.

Group B inspection shall consist of performing the inspections and optical tests specified in table V herein, in the sequence shown.

Group C inspection shall consist of performing the inspections and optical tests specified in table VI herein, in the sequence shown.

TABLE III. Qualification inspection.

Inspection	Sample quantity
Group I	
Visual and mechanical	6 units
Insertion loss	6 units
Group II	
Bandwidth	6 units
Crosstalk	6 units
Group III	
Cable retention	6 units
Temperature cycling	6 units
Storage temperature	3 units
Humidity	2 units
Vibration	2 units
Low pressure (altitude)	2 units

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TABLE IV. Group A inspection.

Inspection
Visual and mechanical
Insertion loss

TABLE V. Group B inspection.

Inspection
Bandwidth
Crosstalk

TABLE VI. Group C inspection.

Inspection	Sample quantity
Subgroup I (3-month interval)	
Cable retention	4 units
Temperature cycling	4 units
Subgroup II (6-month interval)	
Storage temperature	2 units
Humidity	2 units
Vibration	2 units
Subgroup III (12-month interval)	
Low pressure (altitude)	2 units

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Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI, SC
Navy - AS, CG, EC, MC
Air Force - 99
DLA - ES

User activities:
Army - AV
Navy - CG, MC

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6020-0002-4)

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